

FUNDAMENTALS OF COMPUTERS

Chapter – 1 - Computer Basics:

- Introduction,
- Characteristics of computers,
- Evolution of computers,
- Generations of computers,
- Classification of computers,
- The computer system,
- Application of computers

Introduction

The term computer is derived from the word compute, which means to calculate. A computer is an electronic device that processes data based on software instructions. It takes input, processes it using the CPU, stores information, and produces output.

Characteristics of Computers

- Speed
- Accuracy
- Diligence
- Reliability
- Storage capability
- Versatility
- Resource sharing

Speed

- The computers process data at an extremely fast rate, at millions or billions of instructions per second.
- A computer can perform a huge task in a few seconds that otherwise a normal human being may take days or even years to complete.
- The speed of a computer is calculated in MHz (megahertz), GHz (gigahertz) that is, one million instructions per second, one billion instructions per second.
- MHz: A megahertz represents one million cycles per second.
- GHz: A gigahertz represents one billion cycles per second.

Accuracy

- The computers are also very accurate.
- Since the Computer is capable of doing only what it is instructed to do, faulty instructions for data processing may lead to faulty results.
- This is known as Garbage in Garbage Out (GIGO).

Diligence

- Computer, being a machine, does not suffer from the human traits of tiredness and lack of concentration.
- If four million calculations have to be performed then the computer will perform the last four-millionth calculation with the same accuracy and speed as the first calculation.

Reliability

- Generally, reliability is the measurement of the performance of a computer, performing consistently well which is measured against some predetermined standard (specifications) for operation without any failure.
- Computer reliability, in its simplest terms, means your machine works as expected. The computer screen lights up after you press the power button, and the operating system, programs, and apps function as they're supposed to.
- Example: buying performing consistently gadgets

Storage Capability

- Computers can store large amounts of data and can recall the required information almost instantaneously.
- The main memory of the computer is relatively small and it can hold only a certain amount of data: therefore, the data are stored on secondary storage devices such as magnetic tape or disks.
- Small sections of data can be accessed very quickly from these storage devices and brought into the main memory, as and when required, for processing.

Versatility

- Computers are quite versatile in nature. It can perform multiple tasks simultaneously with equal ease.
- For example, at one moment it can be used to draft a letter, another moment it can be used to play music and in between, one can print a document as well.

Resource Sharing

- With the tremendous growth in computer technologies, computers today have the capability to connect with each other. This has made the sharing of costly resources like printers possible.
- Apart from device sharing, data and information can also be shared among groups of computers, thus creating a large information and knowledge base.
- Examples include Connecting to computer, printers, storage devices, software (applications like excel, Anydesk)

Evolution of Computers

- The **Abacus** was probably the first calculating device. It was used to perform simple calculations like addition, subtraction, multiplication and division.
- In **1642**, a French mathematician, **Blaise Pascal**, invented the first (functional automatic calculator) This brass rectangular box, also called a **Pascaline**, used eight movable dials to add numbers up to eight figures long.
- In **1694**, a German mathematician, Gottfried Wilhem von Leibniz, extended Pascal's design to perform multiplication, division and to find square root. This machine is known as the **Stepped Reckoner (Leibniz calculator)**.
- The real beginning of computers as we know them today, however, lay with an English mathematics professor, **Charles Babbage**. In **1822**, he proposed a machine to perform differential equations, called a **Difference Engine**.
- In 1833 he quitted working on this machine to concentrate on the **Analytical Engine**. It had basic element of modern computer like Input, output & memory.

Generation of computer

- The history of computer development is often discussed with reference to the different generation of computer devices. In computer terminology, the word generation is described as a stage of technological development or innovation.
- According to the processor installed in a machine. **There are five generations of computer.**
- First Generation Computer (1940-1956)
- Second Generation Computer (1956-1963)
- Third Generation Computer (1964-1971)
- Fourth Generation Computer(1971-Present)
- Fifth Generation Computer (Present and Beyond)

First generation computer (1940-1956): Vacuum Tubes

- First-generation computers were vacuum tubes/thermionic valve-based machines. These computers used vacuum tubes for circuitry and magnetic drums for memory.
- The input was based on punched cards and paper tape, and the output was in the form of printouts
- First-generation computers relied language also called machine language (language of 0s and 1s) to perform operations and were able to solve only one problem at a time.

Examples: ENIAC, EDVAC and UNIVAC.

ENIAC (Electronic Numerical Integrator and Computer),

EDVAC (Electronic Discrete Variable Automatic Computer),

UNIVAC-1(Universal Automatic Computer-1)

Characteristics of First- Generation Computer

- These were the fastest computing devices of their times (computation time was in milliseconds). Bulky
- These computers were very large and required a **lot of space for installation**.
- Since thousands of vacuum tubes were used (they generated a large amount of heat) Therefore, **air conditioning was essential**.
- These were **non-portable** and very slow equipment's.
- They lacked in **versatility and speed**.
- They were **very expensive** to operate and used a large amount of electricity
- These machines were **unreliable** and prone to frequent hardware failures. Hence, constant maintenance was required.
- Since **machine language** was used, these computers were difficult to program and use
- Each individual component had to be assembled manually. Hence, the commercial appeal of these computers was poor.

Second Generation Computer (1956-1963): Transistors

- In the second generation of computers vacuum tubes are replaced by transistors as the primary electronic component. These are smaller, faster, and more reliable than vacuum tubes.
- It is the ideal replacement which controls heat. This helps to make computers more efficient and durable. Memory: **Magnetic core memory** was used as the primary memory of second-generation computers.
- For secondary storage devices **Magnetic tapes and disks** are used. This helps in more effective storage and faster data access. However, they still relied on punched cards for the input and printouts for the output.
- The development of the generation includes the progress from Machine language to **Assembly language**.
- It uses abbreviations for instructions rather than number. For example: ADD for addition and MULT for multiplication
- High level programming languages such as COBOL and FORTRAN also came into existence in this period.
- Example: PDP-8, IBM 1401 and IBM 7090

Characteristics of Second- Generation Computer

- These machines were based on **transistor technology**.
- These were **smaller** as compared to the first-generation computers.
- The computational time of these computers was reduced to **microseconds from milliseconds**.
- These were more reliable and less prone to hardware failure.
- These were more portable and generated less amount of heat.
- **Assembly language** was used to program computers. Hence, programming became more time-efficient and less cumbersome.

- Second-generation computers still required **air conditioning**.

Third Generation Computer (1956-1963): Integrated Circuits

- The development of the integrated circuit, also called an IC. An IC consists of a single chip (usually silicon) with many components such as transistors and resistors (regulating voltage) fabricated on it.
- ICs replaced several individually wired transistors. This development made computers smaller in size, reliable and efficient
- Instead of punched cards and printouts, interacted with third-generation computers through **keyboards and monitors, and interfaced with the operating system**.
- Examples: NCR 395 and B6500.

Characteristics of Third - Generation Computer

- These computers were based on **IC technology**.
- These were able to reduce the **computational time from microseconds to nanoseconds**
- These were **easily portable** and more reliable than the second-generation computers.
- These devices **consumed less power and generated less heat**. In some cases, air conditioning was still required.
- The size of these computers was smaller as compared to previous-generation computers.
- Since hardware rarely failed, the **maintenance cost was quite low**.
- Extensive use of **high-level languages** became possible.

Fourth Generation Computer (Early 1970's to till date): Microprocessors

- Fourth generation computers, which emerged around 1971, are characterized by the use of **Large-Scale Integration (LSI) technology and Very Large Scale Integration (VLSI) technology**, allowing for the integration of thousands of transistors onto a single chip creating microprocessors that were smaller, faster, and more powerful.
- **(LSI)** refers to **integrated** circuits IC containing thousands of transistors on a single chip.
- **Very Large-Scale Integration (VLSI)** hundreds of thousands to millions (and now billions) of transistors onto a single silicon microchip.
- Example: Apple II, Altair 8800 and CRAY-1

Characteristics of Fourth - Generation Computer

- These computers are **microprocessor-based systems**. Example **Intel Core i7 or AMD Ryzen**
- These are **very small in size**.
- These are the **cheapest** among all the other generation computers.
- The **GUI and pointing devices** enabled users to learn to use the computer quickly.
- **Interconnection of computers** led to better communication and resource sharing.
- These are **portable and quite reliable**.

- These machines generate negligible amount of heat, hence do not require air conditioning.
- Hardware failure is negligible so minimum maintenance is required.
- The **production cost is very low**.

Fifth Generation Computer (Present & Beyond): Artificial Intelligence

- **Fifth generations** of computers were based on **AI (Artificial Intelligence)** technology.
- The process of developing fifth-generation computers is still in the development stage. However, the **expert system** concept is already in use.
- The **expert system** is defined as a computer system that attempts to mimic the thought process and reasoning of experts in specific areas. Example: Healthcare - **CaDet** is an expert system for cancer detection, Weather App, Grammarly AI
- In the fifth generation of computers, **VLSI technology and ULSI (Ultra Large Scale Integration) technology** are used and the speed of these computers is extremely high.
- **ULSI (Ultra Large-Scale Integration)** technology of integrating millions or even billions of transistors onto a single semiconductor chip.

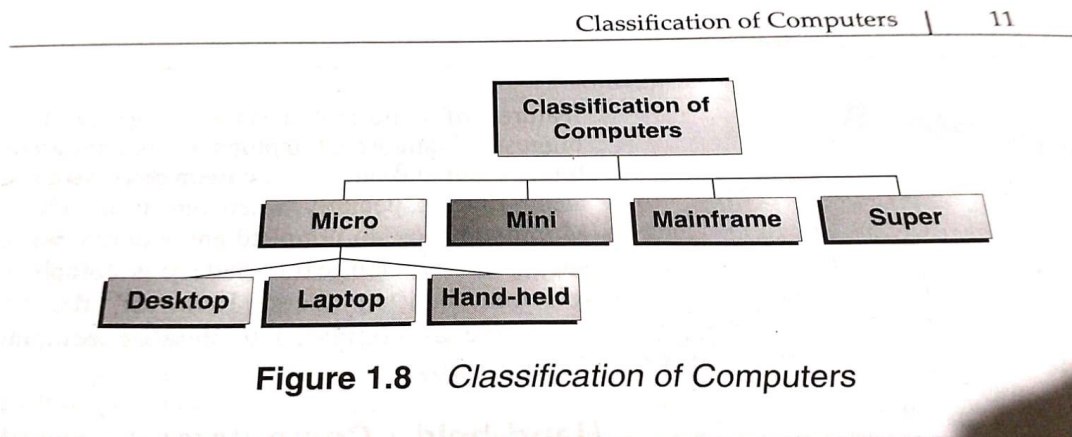
Characteristics of Fifth- Generation Computer

- **Mega Chips:** Fifth-generation computers will use Super Large Scale Integrated (SLSI) chips, were microprocessors having millions of electronic components on single chip, fifth-generation computers require a great amount of storage capacity. Mega chips may enable computer to approximate the memory capacity of humankind.
- **Parallel Processing:** Computers with one processor access and execute only one instruction at a time. This is called serial processing. However, fifth-generation computers will use **multiple processors** and perform parallel processing, thereby accessing several instructions at once and working on them at the same time.
- **Artificial Intelligence (AI):** It refers to a series of related technologies that try to simulate and reproduce human behavior, including thinking, speaking and reasoning. Example: Voice assistants like Amazon Alexa, and technologies like Google Translate, Online Shopping and Recommendations, chatbot, Natural Language Processing.

Classification of Computers

Based on the physical size, performance and application areas, we can generally divide computers into four major categories:

1. Microcomputer,
2. Minicomputer,
3. Mainframe and
4. Supercomputers



1. Microcomputers

- A microcomputer is a small, low-cost digital computer, which usually consists of a microprocessor, a storage unit, an input channel and an output channel, all of which may be on one chip inserted into one or several PC boards.
- An **operating system** and other **software programs** can provide a complete microcomputer system.
- **IBM- PC Pentium 100, IBM-PC Pentium 200 and Apple Macintosh** are some of the examples of microcomputers.
- Microcomputers include **Desktop, Laptop and Hand-held** models such as Personal Digital Assistants (PDAs).

Desktop Computer:

- The desktop computer, also known as the PC, is intended for stand-alone use by an individual. These are the most-common type of microcomputers.
- These microcomputers typically consist of a **system unit**, a **display monitor**, a **keyboard**, an **internal hard disk storage** and other **peripheral devices**.
- The main reason behind the importance of the PCs is that they are **not very expensive** for the individuals or the small businesses.
- Some of the major PC manufacturers are **APPLE, IBM, Dell and Hewlett-Packard**.

Laptop:

- A laptop is a **portable computer**. That a user can carry around. Since the Laptop resembles a notebook, it is also Known as the notebook computer.

- The biggest advantage of laptops is that they are **lightweight** and one can use them anywhere and at any time, especially when one is travelling.
- Moreover, they do **not need any external power supply** as a rechargeable battery is completely self-contained in them.
- However, they are expensive as compared to desktop computers.

Hand-held Computers

- A Handheld computer such as a PDA is a portable computer that can conveniently be stored in a pocket (of sufficient size) and used while the user is holding it.
- A PDA user generally uses a pen or electronic stylus, instead of a keyboard for input
- The monitor is the only apparent form of output.
- Hand-held computers usually have no disk drive; rather, they use small cards to store programs and data.
- They have limited memory and are less powerful as compared to desktop computers.
- Some examples of hand-held computers are Apple Newton, Casio Cassiopeia and Franklin eBookMan.
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2. Mini Computer

- A minicomputer is a small digital computer, which normally is able to **process and store less data than a mainframe** but **more than a microcomputer**.
- Generally, these computers are used as desktop devices that are **often connected to a mainframe** in order to perform the auxiliary operations.
- It is capable of supporting from four to about 200 simultaneous users.
- It **serves as a centralized storehouse** for cluster of workstations or as a network server. Minicomputers are usually **multi-user systems** so these are used in interactive applications in industries, research organizations, colleges and universities.
- Some of the widely used minicomputers are **PDP 11, IBM (8000 series) and VAX 7500**.

3. Mainframes:

- A mainframe is an **ultra-high-performance** Computer made for high-volume, **processor- Intensive computing** (heavily rely on CPU). It consists of a high-end Computer processor, with related peripheral
- Normally, it is able to process and store more data than a minicomputer and far more than a microcomputer,
- Mainframes are the second Largest (in capability and size) of the computer family. The largest being the supercomputers.
- Mainframes can usually execute many programs simultaneously at a high speed, whereas supercomputers are designed for a Single process. Example: Bank's mainframe processing numerous transactions from ATMs and online banking in real-time.
- It is typically used by large businesses and for scientific purposes. **Some examples of the mainframe are IBM's ES000, VAX 8000 and CDC 6600.**

4. Supercomputers

- Supercomputers are the special-purpose machines, which are especially designed to **maximize the numbers of floating-point operations per second (FLOPS)**.
- A supercomputer can process a great deal of data and make extensive complex mathematical equations very quickly, which would have taken many years when performed using a paper and pencil or using a hand calculator.
- They are also required by the military strategists to simulate defence scenarios. Cinematic specialists use them to produce sophisticated movie animations.
- The largest commercial uses of supercomputers are in the entertainment/advertising industry. CRAY-3, Cyber 205 and PARAM are some well-known supercomputers.

THE COMPUTER SYSTEM

- A computer can be viewed as a system, which consists of a number of interrelated components (CPU, RAM, input device, output device) that work together with the aim of converting **data into information**.
- The set of instructions given to computers are called **programs**. Without programs, computers would be useless. Examples web browsers like Chrome or Firefox, word processors like Microsoft Word, and games like Minecraft
- The physical parts that make up a computer (the CPU, input, output and storage unit) are known as **hardware**.
- Any hardware device connected to the computer or any part of the computer outside the CPU and working memory is known as a **peripheral**.
- Some examples of peripherals are keyboards, mouse and monitors.

Components of a Computer System

- There are several computer systems in the market with a wide variety of makes, models and peripherals. In general, a computer system comprises the following components
- **CPU:** This unit performs processing of instructions and data inside the computer.
- **Input Unit:** This unit accepts instructions and data.
- **Output Unit:** This unit communicates the results to the user.
- **Storage Unit:** This unit stores temporary and final results

Central Processing Unit (CPU):

- Central Processing Unit: The CPU, also known as a processor, is the brain of the computer system that processes data (input) and converts it into meaningful information (output).
- The CPU works with data in discrete form, that is, either 1 or 0.
- Eventually, the results of these operations are translated into characters, numbers and symbols that can be understood by the user.

The CPU itself has three parts:

- **Arithmetic Logic Unit (ALU),**
- **Control Unit,**
- **Registers**

Arithmetic Logic Unit (ALU): This unit performs the arithmetic (add, subtract, multiplication, division) and logical operations (AND, OR, NOT) on the data made available to it.

Control Unit: This unit checks the correctness of the sequence of operations. It fetches the program instructions from the memory unit, interprets them and ensures correct execution of the program. It also controls the I/O devices.

Registers: Registers hold operands or instructions that CPU would be currently processing. These are high-speed temporary memory units that can hold varied information such as data, instructions, and intermediate results of calculations. These are high-speed temporary memory units that can hold varied information such as data, instructions, and intermediate results of calculations.

The registers can be considered as the CPU's working memory.

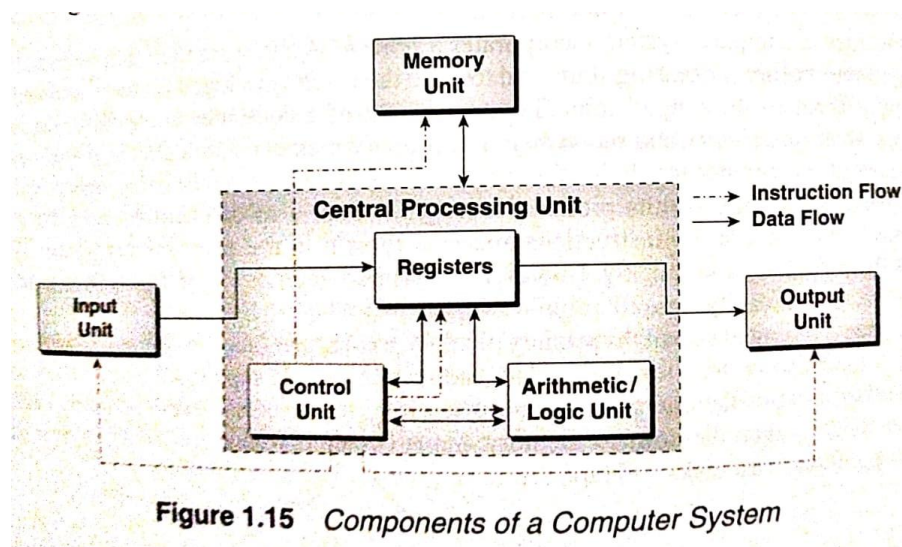


Figure 1.15 Components of a Computer System

Input, Output and Storage Unit

- **An input unit** accepts instructions and data from the user with the help of input devices such as keyboard, mouse, light pen, etc.
- **The output unit** performs just opposite to that of input unit. It accepts the output (which is in machine-coded form) produced by the computer, converts them into the user understandable form and supplies the converted results to the user with the help of an output device such as printer, monitor and plotter.
- **The storage unit** of a computer comprises two types of memory/storage: **primary and secondary**.

The primary memory, also called the main memory, is the part of a computer that holds the instructions and data currently being processed by the CPU, the intermediate results.

Secondary memory which is extensively used for storing data and instructions. Some commonly used secondary memory are floppy disks, hard disks and type drives.

Applications of computers

- Science
- Education
- Medicine and Healthcare
- Engineering/Architecture/Manufacturing
- Entertainment
- Communication
- Business Application
- Banking

Science:

- Scientists have been using computers to develop theories, to analyze and to test the data.
 - They can be used to generate detailed studies of how earthquakes affect buildings or pollution affects weather pattern.
 - Satellite-based applications would not have been possible without the use of computers.
- Example: GPS, weather forecasting.

Education:

- Computers are very crucial for online classes, download study material on the internet and interact with fellow learners.
- Computers are also used to track student attendance and learning strategies.
- Currently, the classrooms, libraries and museums are utilizing computers to make the education much more interesting.

Medicine and Healthcare:

- Computers are used to record patients' information, monitoring heart rate, oxygen level, and blood pressure.
- By using automated imaging techniques, doctors can look inside a person's body and can study each organ in detail (such as CAT scans or MRI scans), which was not possible few years ago.
- Now, doctors are using computers right from diagnosing the illness to monitoring a patient's status during complex surgery.

Engineering/Architecture/Manufacturing:

- The architects and engineers are extensively using computers in designing and drawings. Computers can create objects that can be viewed from all the three dimensions (3D).
- The manufacturing factories are using computerized robotic arms in order to perform hazardous jobs.

Entertainment:

- We can play various interesting video games using a computer. We can watch movies, TV shows, and reality shows on the computer.
- The special effects, which mesmerize the audience, would not have been possible without the computers. In addition, computerized animation and colourful graphics have modernized the film industry.

Communication:

- E-mail or electronic mail is one of the communication media in which computers are used.
- Through an e-mail, messages and reports are passed from one person to one or more persons with the aid of computers and telephone lines.
- The advantage of this service is that while transferring the messages it saves time, avoids wastage of paper, and so on.

Banking:

- In the field of banking and finance, computers are extensively used.
- People can use the Automated Teller Machine (ATM) services 24 hours a day in order to deposit and withdraw cash.
- When the different branches of the bank are connected through the computer networks, the inter-branch transactions, such as drawing cheques and making drafts, can be performed by the computers without any delay.